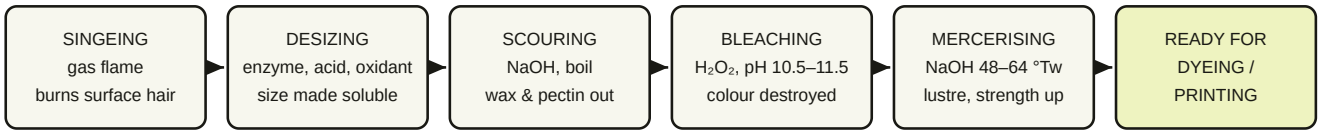
TEXORA · LECTURE SHEET

Preparatory Wet Processes: Singeing, Desizing, Scouring, Bleaching and Mercerization

The chemistry and technology of the five core preparation operations — burning off surface hair, stripping the size film, boiling out natural impurities in alkali, oxidising away colour, and swelling cotton in concentrated caustic soda — plus shearing, enzymatic pretreatment, and how each fibre family is prepared for coloration.

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 4 · LEVEL 2 TERM 2

1. The sequence in process-flow form



Hot washing follows desizing, scouring and bleaching; mercerising is applied to scoured (usually bleached) goods

Fig. 4 — Cotton preparation flow with the working chemistry of each stage

Fig. 4 — Process flow of the core preparatory operations for cotton.

2. Singeing

Object: to burn off the protruding fibre ends (the "hairiness") standing out of the fabric surface, giving a smooth, clean surface that prints sharply, shows fewer specks and pills less. Only the surface hairs burn — the fabric passes so fast that the body of the cloth does not heat through.

- **Gas singeing** (standard): fabric passes at 50–150 m/min over one or more rows of gas flames, both faces as required; immediate quenching in a water/desize bath follows to kill sparks.
- **Plate singeing:** fabric glides over red-hot curved copper plates — older, less uniform.
- **Rotary-cylinder singeing:** fabric contacts a heated rotating cylinder — used for some velvets and where gas is unavailable.

Over-singeing scorches and yellows the fabric; under-singeing leaves hairiness that traps dye and prints fuzzy outlines. Thermoplastic fabrics need care — polyester surface fibres can melt into hard beads

instead of burning cleanly, so singeing is gentler or omitted.

3. Desizing

Object: complete removal of the warp size applied before weaving (8–15% on warp weight for cotton). The size must be degraded to water-soluble fragments, then washed out before any other bath touches the fabric. Method follows size chemistry:

METHOD	AGENT / CONDITIONS	ACTION AND NOTES
Enzymatic	α-Amylase, pH ≈ 5.5–7.5, 60–80 °C (thermostable grades higher)	Hydrolyses starch to soluble dextrins and sugars; the standard for starch size; does not attack PVA
Acid steeping	Dilute H_2SO_4 (≈ 5–10 g/L), room temperature, hours of storage	Hydrolyses starch; risk of tendering cotton if over-long or over-strong
Rot steeping	Soaking in water 24–48 h	Bacterial fermentation degrades starch; slow, uneven, malodorous — obsolete except small scale
Oxidative	Persulphate, bromite, peroxide-based	Breaks down starch *and* synthetic sizes; can be combined with scouring
Hot washing only	Hot water/detergent wash	Sufficient for fully water-soluble sizes (PVA, CMC, polyacrylates)

Completeness is judged later by absorbency and by the iodine test for residual starch (Topic 6). A fabric desized but not washed merely re-deposits dextrins as it dries.

4. Scouring

Object: removal of the natural impurities of the fibre — for cotton, waxes, pectins, proteins, hemicelluloses and mineral matter — so the fabric becomes absorbent. The engine is hot caustic soda:

- **Saponification:** the fatty part of cotton wax (fatty acids and saponifiable esters) is converted to soluble soap, which itself helps emulsify the rest.

$$\text{R–COOH} + \text{NaOH} \rightarrow \text{R–COONa (soap)} + H_2O$$

- **Emulsification:** non-saponifiable wax (higher alcohols, hydrocarbons) is emulsified by the generated soap plus added surfactant and held in the bath.

- **Hydrolysis/solubilisation:** pectins are hydrolysed to soluble pectates; proteins to soluble amino-acid salts; hemicelluloses dissolve in the alkali; the fibre swells, loosening seed-coat fragments so washing expels them.

Typical cotton conditions: NaOH 2–5% owf (or 5–10 g/L in continuous pad-steam), wetting agent ≈ 0.5–1 g/L, sequestering agent; kier boiling at 100–130 °C under pressure for 4–8 h batchwise, or pad-steam at 100–102 °C for 1–3 minutes continuously (with stronger alkali). Scouring loss of cotton is ≈ 4–8% of fabric weight — the mass of impurities leaving. **Bio-scouring** with pectinase enzymes works at 50–60 °C near-neutral pH, removing pectin (the "glue" binding wax to fibre) so wax washes out — less weight loss, softer handle, lower effluent load, but wax removal is less total than alkali scouring. Fabric for dark shades may be scoured only; white and pastel goods continue to bleaching.

5. Bleaching

Object: destruction of the natural colouring matter (flavone pigments in cotton, etc.) to a permanent, level white without significant fibre damage. Bleaching is controlled oxidation (or, for some synthetics/wool, reduction) of chromophores; the three working agents:

AGENT	CONDITIONS	ADVANTAGES / LIMITS
Hydrogen peroxide H_2O_2	pH 10.5–11.5, 80–95 °C, stabiliser (silicate/organic); active species is the perhydroxyl ion HOO^-	Universal (cotton, blends, wool with care); no harmful residues, no AOX; needs stabilisation against metal-catalysed decomposition
Sodium hypochlorite NaOCl	pH 9–11, room temperature to 30 °C; active species $HOCl/OCl^-$	Cheap and fast on cotton; forms chlorinated by-products (AOX), can tender and yellow if pH falls; largely phased out
Sodium chlorite $NaClO_2$	pH 3.5–4.5, 80–95 °C, activator; active species ClO_2	Excellent whiteness with little damage, good for synthetics/blends; corrosive ClO_2 fumes demand enclosed plant

$$H_2O_2 + OH^- \rightleftharpoons HOO^- + H_2O$$

Uncontrolled peroxide decomposition ($2\ H_2O_2 \rightarrow 2\ H_2O + O_2$) wastes bleach and, when metal-catalysed at a point, digs pinholes — hence stabilisers and sequestering agents (Topics 2–3) and an **anti-chlor/peroxide clean-up** (catalase enzyme removes residual H_2O_2 before dyeing). Over-bleaching any cellulosic produces **oxycellulose** — tender, alkali-sensitive fibre that dyes unevenly; bleaching is always a compromise between whiteness and damage.

6. Mercerization

Fabric or yarn is treated under tension with concentrated caustic soda, **18–25% NaOH (≈ 48–64 °Tw)**, at room temperature for 30–60 s, then washed out under tension. The alkali swells the cotton fibre from a flattened kidney-bean cross-section to an almost round one; the convolutions disappear. Consequences:

- **Lustre:** the smooth, round surface reflects light regularly — the signature mercerised shine.
- **Strength:** increases by roughly 10–20%.
- **Dye affinity:** more accessible internal surface — dye uptake rises markedly (a mercerised fabric dyes visibly deeper with the same recipe; recipes must be adjusted).
- **Dimensional stability and absorbency** also improve. Degree of mercerisation is checked by the barium activity number (Topic 6). Without tension the same treatment is **causticisation** (slack mercerisation), used for stretch and crimp effects.

7. Shearing and enzymatic pretreatment

Shearing (cropping) mechanically trims the surface with spiral blades to a uniform pile height — the mechanical complement of singeing for knitted and raised goods. **Enzymatic pretreatments** apply the specificity of enzymes: amylase desizing (Section 3), pectinase bio-scouring (Section 4), and **cellulase biopolishing**, which hydrolyses the micro-fibrils ("fuzz") on cotton/knit surfaces to give a smoother, clearer-printing fabric with less pilling — at the cost of a few percent weight and strength if overdosed.

8. Preparation of each fibre for coloration

MATERIAL	PREPARATION ESSENTIALS
Cellulose (cotton, linen)	Full sequence: desize → scour (hot NaOH) → peroxide bleach → (mercerise). Fibre tolerates alkali and oxidant within controlled limits.
Regenerated cellulose (viscose)	Gentler version of the cotton sequence: lower alkali, lower temperature, minimal tension — wet viscose is weak and swells strongly.
Silk	Degumming: sericin removed by soap solution near the boil (or enzymes), losing 20–30% weight; then mild peroxide bleaching. Alkali and chlorine bleach damage fibroin.
Wool	Scouring with mild detergent/soap at ≤ 50–55 °C (removes lanolin and suint) with minimal agitation to avoid felting; carbonising with H_2SO_4 bakes out vegetable matter; bleaching with peroxide (or reductive bleach); chlorination only as a controlled anti-shrink treatment.

Jute	Emulsion batching (oil + water) and 24–48 h conditioning to soften; mild scouring; careful peroxide bleaching — lignin-rich fibre yellows and weakens under aggressive bleach or alkali.
Synthetics	Scouring only: removal of spin finish and oils with detergent (polyester may need an alkaline reduction-clean later); heat setting locks dimensions; most need no bleaching.
Blends	Union preparation set by the most sensitive component: polyester/cotton follows the cotton sequence with peroxide (safe for both); wool blends exclude strong alkali and chlorine entirely.

9. Glossary — key terms

- **Singeing:** burning protruding surface fibres with gas flame for a smooth fabric face.
- **Saponification:** conversion of fats/wax esters to soluble soap by alkali — the core scouring reaction.
- **Oxycellulose:** oxidatively damaged cellulose from over-bleaching; tender and dye-resisting.
- **Mercerisation:** treatment of cotton under tension with concentrated NaOH, giving lustre, strength and dye affinity.
- **Degumming:** removal of sericin from silk.
- **Carbonising:** acid baking of vegetable matter out of wool.

10. Key points

- Order matters: singe → desize → scour → bleach → mercerise, washing between stages.
- Scouring chemistry = saponification + emulsification + hydrolysis in hot NaOH; cotton loses ≈ 4–8% weight.
- Peroxide (HOO^- at pH 10.5–11.5) is the universal bleach; hypochlorite is being phased out (AOX); over-bleaching makes oxycellulose.
- Mercerising: 48–64 °Tw NaOH under tension rounds the fibre — lustre, +strength, +dye uptake.
- Every fibre is prepared by the mildest route its impurities allow: viscose gently, wool without hot alkali, silk by degumming, synthetics by scouring alone.

11. Summary

The five core processes form one continuous argument. Singeing clears the surface mechanically, desizing dissolves the weaver's film, scouring boils out the fibre's own waxes and pectins through

saponification and emulsification, bleaching oxidises the remaining colour bodies under stabilised control, and mercerisation permanently remodels the cotton fibre in strong caustic soda. Enzyme alternatives soften several of these steps. Adapted in severity to the fibre — full force for cotton, caution for viscose, wool, silk and jute, minimal for synthetics — the sequence delivers the clean, white, absorbent substrate that coloration requires.

12. Practice

Answer in your own words; brief model answers follow.

1. A fabric is bleached before it is scoured. Give two reasons the result will be poor.
2. Write the saponification reaction for a generic wax fatty acid and explain how the product assists removal of the non-saponifiable wax fraction.
3. Calculate the approximate NaOH strength in % w/v of a mercerising liquor of 54 °Tw, given °Tw = 200 × (specific gravity − 1) and a liquor of sp. gr. 1.27 contains about 315 g NaOH per litre. (State each step.)
4. Why is sodium hypochlorite being replaced by hydrogen peroxide in modern cotton bleaching? Give three reasons.
5. State the preparation route for a wool fabric, naming the step that removes vegetable matter and the two process variables most tightly controlled to protect the fibre.

Answer key

1. (1) Bleach is wasted on size film and surface soil instead of fibre pigments, and the still-present wax blocks uniform bleach uptake, giving patchy whiteness. (2) Once bleached, the fabric still has to be scoured — hot alkali after bleaching attacks the already-oxidised fibre and yellows the white. Scour first, then bleach the clean fibre.
2. R–COOH + NaOH → R–COONa + H_2O. The soap formed is itself a surfactant: it emulsifies the non-saponifiable waxes (higher alcohols, hydrocarbons) into the bath, so both wax fractions leave together.
3. 54 °Tw → sp. gr. = 1 + 54/200 = 1.27. From the table value, 1 litre contains ≈ 315 g NaOH, i.e. ≈ 31.5% w/v — inside the usual mercerising range (18–25% w/w ≈ 48–64 °Tw liquors) and consistent with commercial practice.
4. (1) Hypochlorite forms chlorinated organic by-products (AOX) restricted by effluent rules and ZDHC; peroxide leaves only water and oxygen. (2) Hypochlorite can tender and yellow cotton if pH/temperature control slips, and residual chlorine continues attacking fibre in storage. (3) Peroxide suits all fibres and blends and combines with scouring in one bath, saving water and energy.

5. Mild detergent scouring (≤ 50–55 °C) → carbonising with sulphuric acid to bake out vegetable matter → gentle peroxide bleach → dye. The tight controls are temperature/alkali (hot strong alkali dissolves and felts wool) and mechanical agitation (drives felting shrinkage).
